Data Quality Basics

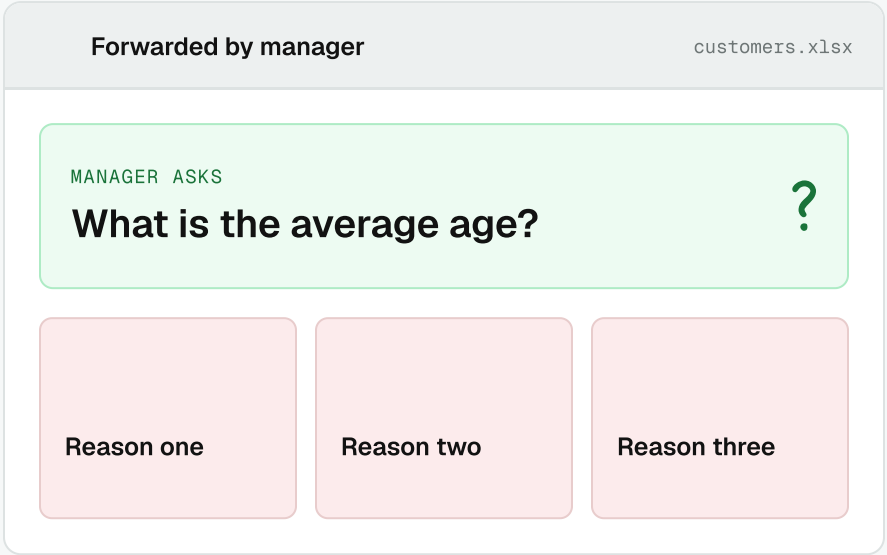
BY THE END

You can inspect a table and name the quality problems in it.

Overview

A manager forwards you a customer list and asks for the average age. Two minutes in, you have found three separate reasons the answer would be wrong.

Almost every analysis you will ever run begins with a moment like this. Bad data leads to bad decisions. Even simple analysis can fail if the data is incomplete, incorrect, or inconsistent.

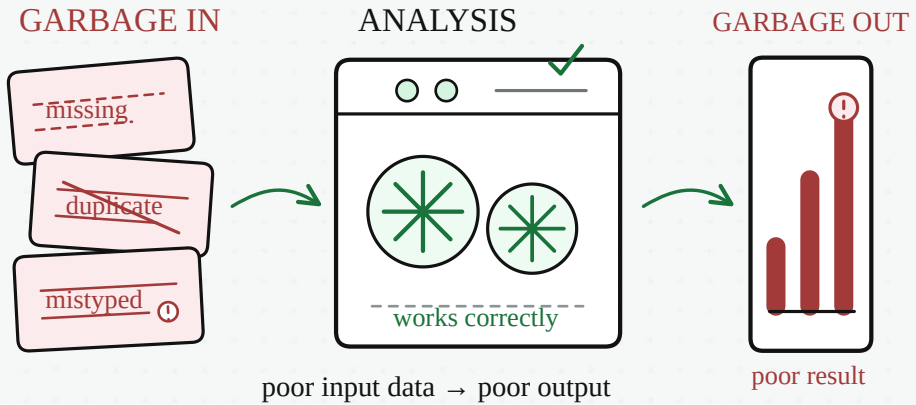


Why data quality matters

Data quality tells you whether a dataset is fit for the job you need it to do. A decision built on missing, duplicated, or mistyped numbers can look completely logical and still be wrong, because nothing in the reasoning shows the fault.

> **It is in the inputs.**
> This idea is often summarized as **garbage in, garbage out (GIGO)**: poor input data produces poor output, even when the calculation or tool works correctly.

That is why data professionals spend significant time checking data before analyzing it.



Common data quality problems

Some of the most common issues include:

Missing values

blank cells or empty fields

Duplicates

the same record appearing more than once

Incorrect values

typos, wrong units, or impossible numbers

Inconsistent formatting

dates written as `01/05/2024` and `2024-05-01` in the same dataset

Outdated data

information that no longer reflects the current situation

None of this is unusual. Expect to find several of them in any dataset you are handed.

Key quality dimensions

DIMENSION	QUESTION IT ANSWERS
Accuracy	Is the data correct?
Completeness	Is anything missing?
Consistency	Is the data formatted the same way throughout?
Timeliness	Is the data up to date?
Relevance	Does the data actually help answer the question?

You do not need advanced tools to start checking quality. Sometimes a simple review of a table can reveal missing values, strange entries, or duplicate rows.

EXAMPLE

Find the four problems

read before going on

This is the customer list from the top of the lesson. The manager wants the average age. Read it before going on. There are four problems in it.

CUSTOMER ID	NAME	AGE	CITY	EMAIL
C-101	Maya Chen	34	Springfield	maya@example.com
C-102	Noah Reed	27	Riverton	noah@example.com
C-103	Elena Cruz		springfield	elena@example.com
C-104	Liam Brown	200	Riverton	liam@example.com
C-102	Noah Reed	27	Riverton	noah@example.com

Three problems break the average

Three of the four break the average age, and each one breaks it differently:

INCORRECT

Liam Brown	200

Liam's age is 200. It is **inaccurate**, and a single impossible value drags the average up on its own.

INCOMPLETE

Elena Cruz	blank

Elena's age is blank. That is **incomplete** - and you must decide whether she is left out of the average or counted as zero, which give very different answers.

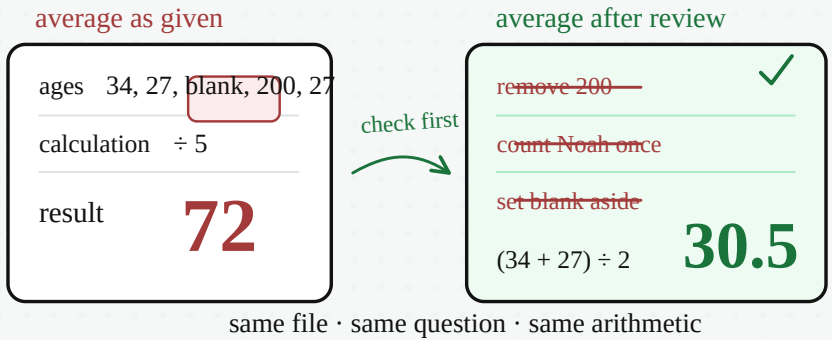
DUPLICATE

C-102 · Noah	27
C-102 · Noah	27

C-102 appears twice, and every value in it is correct. That is the point: nothing in either row is wrong, but Noah is counted as two customers, so his age carries double the weight of anyone else's.

Same file, different answer

Average the ages exactly as the table gives them and you get **72**. Remove the impossible value, count Noah once, and set Elena's blank aside, and you get **30.5**. Same file, same question, same arithmetic, and an answer that would have sent the manager looking for a customer base that does not exist.



`Springfield`

`springfield`

The fourth does not touch the average at all. `Springfield` and `springfield` are **inconsistent**, and it would matter enormously if the question were about cities. But the manager asked about age. Which faults matter depends on the question you were asked.